NEXORA TECHNOLOGIES PVT. LTD.

Cloud, Infrastructure, IT & Enterprise Technology Handbook

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1. Purpose, Scope & Operating Model

a. Purpose

This handbook establishes the standards and operating practices used by Nexora Technologies for cloud platforms, infrastructure, enterprise technology and internal IT services.

It defines how Nexora provisions, operates, secures, monitors and supports technology infrastructure used by employees, engineering teams, business functions and customer-facing workloads.

The handbook covers:

- Cloud platforms
- Infrastructure engineering
- Platform engineering
- Networking
- Compute
- Storage
- Databases
- Identity infrastructure
- Corporate IT
- Employee devices
- SaaS applications
- Service Desk
- IT asset management
- Monitoring
- Observability
- Backup
- Recovery

- Capacity management
- Infrastructure security
- Cloud cost management
- Infrastructure lifecycle

It applies to company-managed infrastructure and approved third-party technology environments.

b. Operating Principles

Nexora infrastructure is operated according to the following principles:

I. Standardize where practical

Common platforms, configurations and operational processes should be reused wherever they meet business requirements.

II. Automate infrastructure

Infrastructure provisioning and repeatable operational activities should use automation where practical.

III. Secure by default

Infrastructure should use appropriate identity, network, encryption and monitoring controls before production use.

IV. Minimize unnecessary access

Infrastructure access follows least-privilege principles.

V. Monitor what matters

Critical systems must have sufficient observability to detect failures and abnormal behavior.

VI. Design for recovery

Important workloads must have appropriate backup and recovery capabilities.

VII. Manage cost deliberately

Cloud and technology consumption must be visible, monitored and aligned with business value.

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2. Infrastructure Organization

a. Infrastructure Functions

The Cloud & Infrastructure organization contains several specialized capabilities.

Function	Primary Responsibility
Cloud Engineering	Cloud architecture and services
Platform Engineering	Internal developer platforms

Function	Primary Responsibility
Site Reliability Engineering	Reliability and production operations
Network Engineering	Corporate and cloud networking
Database Engineering	Database platforms
Infrastructure Security	Infrastructure security controls
Corporate IT	Employee technology
IT Service Management	Service Desk and IT processes
IT Asset Management	Hardware and software assets
Cloud FinOps	Cloud cost governance

b. Geographic Operations

Infrastructure support operates across Nexora's principal locations:

- Bengaluru
- Pune
- Hyderabad
- London

Critical infrastructure support is organized around shared enterprise platforms rather than individual offices.

c. Infrastructure Ownership

Every production infrastructure component should have an identifiable owner.

Ownership should include:

- Technical team
- Service owner
- Business owner where appropriate
- Environment
- Support path
- Escalation path

Unowned infrastructure is considered an operational risk.

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3. Enterprise Technology Architecture

a. Technology Layers

Nexora's technology environment can be represented in six major layers.

```
Business Applications
        ↓
Application Platforms
        ↓
Data & Integration
        ↓
Cloud & Compute
        ↓
Network & Identity
        ↓
Physical / Endpoint Infrastructure
```

Each layer may be operated by different technical teams.

b. Enterprise Technology Platforms

Major internal platforms include:

Platform	Primary Purpose
Nexora PeopleHub	Human resources
Nexora ServiceDesk	IT service management
Nexora Workspace	Collaboration
Nexora Identity	Identity and access
Nexora KnowledgeHub	Internal knowledge
Nexora Insight	Enterprise analytics
Nexora Flow	Workflow automation
Nexora Sentinel	Security monitoring
Nexora Assist	Enterprise AI

c. Standard Cloud Provider

Nexora's primary enterprise cloud platform for cloud-native workloads is **Google Cloud Platform (GCP)**.

Other cloud platforms may be used when:

- Required by a customer
- Required by an acquired technology
- Required by a specific technical workload
- Required for geographic or regulatory reasons

Multi-cloud usage must have an identified business or technical rationale.

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4. Cloud Account & Project Structure

a. Cloud Organization

Cloud resources are organized using controlled organizational structures.

A typical hierarchy is:

```
Nexora Cloud Organization
        │
        ├── Production
        │     ├── Customer Workloads
        │     ├── Enterprise Platforms
        │     └── Shared Services
        │
        ├── Non-Production
        │     ├── Development
        │     ├── Testing
        │     └── Staging
        │
        └── Security & Governance
```

b. Project Naming

Cloud projects should use consistent naming.

Example:

```
NXR-DATA-PROD-01
NXR-DATA-STG-01
NXR-PLATFORM-PROD-01
NXR-SECURITY-PROD-01
NXR-DEVTOOLS-NONPROD-01
```

c. Resource Ownership

Each production cloud project should identify:

- Business owner
- Technical owner
- Environment
- Cost center
- Data classification
- Support team

d. Project Creation

New production projects require approval from the relevant platform owner.

Projects should not be created solely for convenience when an existing approved project can satisfy the requirement.

5. Cloud Environment Standards

a. Environment Model

Nexora uses separate environments for different lifecycle stages.

Environment	Purpose
Development	Engineering development
Test	Automated and functional testing
Staging	Production-like validation
Production	Live business workloads

Some teams may combine development and test environments for low-risk applications.

b. Production Isolation

Production environments must be logically separated from development environments.

Production credentials must not be reused in non-production environments.

c. Production Data

Production data must not be copied into development environments without an approved data-handling process.

Where test data is required, teams should use:

- Synthetic data
- Masked data
- Anonymized data
- Approved test datasets

6. Compute Infrastructure

a. Compute Options

Nexora may use:

- Virtual machines
- Containers
- Managed container platforms
- Serverless compute
- Batch processing

- Specialized compute

Technology selection depends on:

- Workload
- Scale
- Operational requirements
- Cost
- Security
- Team capability

b. Virtual Machines

Virtual machines should use approved operating-system images.

Production VM configurations should be documented.

c. Containers

Containerized workloads should:

- Use approved base images
- Be vulnerability scanned
- Avoid running with unnecessary privileges
- Avoid embedded credentials
- Define resource requirements
- Use versioned images

d. Resource Allocation

Production services should define:

- CPU requirements
- Memory requirements
- Storage requirements
- Scaling limits

Teams should avoid unlimited resource allocation.

---

7. Kubernetes & Container Platforms

a. Container Orchestration

Nexora may use managed Kubernetes or equivalent approved container platforms for workloads requiring orchestration.

Typical use cases include:

- Microservices
- APIs
- Background workers
- Data services
- Internal platforms

b. Cluster Standards

Production clusters should have:

- Centralized authentication
- Network controls
- Resource limits
- Logging
- Monitoring
- Vulnerability scanning
- Backup strategy where applicable

c. Namespace Management

Namespaces should be separated by:

- Environment
- Application
- Business function

Teams should avoid placing unrelated production workloads in a single uncontrolled namespace.

d. Resource Limits

Production workloads should define CPU and memory requests and limits where supported.

A service repeatedly exceeding **80% of its allocated memory** should be reviewed for optimization or capacity adjustment.

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8. Networking

a. Network Architecture

Nexora network architecture separates:

- Corporate networks
- Cloud networks
- Production workloads
- Non-production workloads
- Security services

- External integrations

b. Virtual Networks

Production cloud workloads should use approved virtual network configurations.

Network segmentation should reflect workload sensitivity.

c. Subnets

Subnets may be separated by:

- Environment
- Application tier
- Region
- Security boundary

d. Firewall Rules

Firewall rules should follow least privilege.

Rules should specify:

- Source
- Destination
- Port
- Protocol
- Purpose
- Owner

Broad rules such as unrestricted access from the public internet should be avoided unless explicitly required and approved.

e. Network Changes

Material firewall and routing changes must use approved change-management procedures.

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9. Internet Connectivity

a. Public Endpoints

Public-facing services must be reviewed for:

- Authentication
- TLS
- Firewall controls
- Rate limiting

- Logging
- Monitoring
- DDoS protection where applicable

b. TLS

External services should use current approved TLS configurations.

Expired or invalid certificates must be remediated promptly.

c. DNS

Corporate and production DNS zones must be administered through approved services.

DNS changes affecting production services require appropriate authorization.

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10. Identity & Access Management

a. Identity Principles

Access follows:

- Least privilege
- Need to know
- Role-based access
- Strong authentication
- Periodic review

b. Nexora Identity

Nexora Identity is the central enterprise identity platform.

It supports:

- Employee authentication
- Multi-factor authentication
- Role assignment
- Application access
- Access lifecycle
- Privileged access controls

c. Multi-Factor Authentication

MFA is mandatory for:

- Corporate accounts
- Cloud consoles

- Privileged access
- Security platforms
- Sensitive business systems

d. Privileged Access

Privileged access should be:

- Limited
- Approved
- Logged
- Reviewed

Permanent administrative access should be minimized.

e. Access Reviews

Critical application and infrastructure access should normally be reviewed at least **quarterly**.

Privileged access should receive enhanced review.

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11. Joiner, Mover & Leaver Technology Access

a. Joiners

When an employee joins, Corporate IT provisions:

- Identity
- Email
- Collaboration tools
- Required applications
- Device
- Role-specific access

Standard equipment should be ready by the employee's first working day.

b. Movers

When an employee changes role:

- Existing access should be reviewed
- New access should be approved
- Unnecessary access should be removed

c. Leavers

When employment ends:

- Corporate identity is disabled
- Sessions are revoked where supported
- Device access is removed
- SaaS access is revoked
- Privileged access is removed

Critical access should be disabled promptly according to the approved separation process.

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12. Corporate IT

a. Corporate IT Responsibilities

Corporate IT supports:

- Employee devices
- Corporate applications
- Collaboration
- Connectivity
- Printing where applicable
- Endpoint security
- Software provisioning
- IT support

b. Standard Employee Equipment

Standard equipment may include:

- Business laptop
- Docking station
- Monitor
- Keyboard and mouse
- Headset

Equipment depends on job role.

Engineering, design and data-intensive roles may receive higher-performance configurations.

c. Device Standards

Company-managed devices should have:

- Disk encryption
- Endpoint security

- Screen lock
- Automatic updates
- Central management
- Approved software

d. Device Replacement

Standard corporate laptops should generally be reviewed for replacement after approximately **36 months**, subject to condition, performance and business requirements.

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13. IT Asset Management

a. Asset Inventory

Corporate IT maintains an inventory of managed assets.

Assets may include:

- Laptops
- Desktops
- Monitors
- Mobile devices
- Network equipment
- Servers
- Software licenses

b. Asset Identification

Assets should have unique identifiers.

Example:

```
NXR-LAP-042817
NXR-MON-018274
NXR-MOB-006391
```

c. Asset Lifecycle

```
Procurement
   ↓
Receipt
   ↓
Assignment
   ↓
Maintenance
   ↓
Reassignment
   ↓
Return
   ↓
Secure Disposal
```

d. Lost Equipment

Lost or stolen company equipment must be reported immediately.

Corporate IT and Security may remotely lock or wipe managed devices where supported.

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14. IT Service Desk

a. ServiceDesk

Nexora ServiceDesk is the central system for employee IT requests and incidents.

Employees should create tickets for:

- Device problems
- Application problems
- Access requests
- Network problems
- Software installation
- Hardware replacement

b. Ticket Priority

Priority	Typical Impact
P1	Critical enterprise impact
P2	Significant business impact
P3	Individual or limited impact
P4	General request / low urgency

c. Reference Response Targets

Priority	Initial Response	Target Resolution
P1	15 minutes	4 hours
P2	30 minutes	8 hours
P3	4 business hours	2 business days
P4	1 business day	5 business days

These are standard internal targets and may vary by service.

d. Escalation

Tickets should be escalated when:

- The SLA is at risk
- Multiple employees are affected
- Security is suspected
- A critical business function is unavailable

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15. Enterprise SaaS Applications

a. SaaS Management

Enterprise SaaS applications must have:

- Business owner
- Technical owner
- Contract owner
- Security assessment where applicable
- Cost center
- Access model

b. Application Procurement

Employees must not independently purchase enterprise SaaS using company funds without following procurement procedures.

c. SaaS Access

Access should be provisioned according to role.

Inactive accounts should be reviewed and removed.

d. Application Review

Critical SaaS platforms should be reviewed periodically for:

- Security
- Cost
- Usage
- Vendor performance
- Business value

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16. Storage

a. Storage Types

Nexora may use:

- Object storage
- Block storage
- File storage
- Database storage
- Archive storage

b. Data Classification

Storage controls should reflect data classification.

Restricted information requires stronger controls than Public information.

c. Object Storage

Object storage buckets should not be publicly accessible unless there is an approved business requirement.

d. Lifecycle Management

Storage lifecycle rules should move or delete data according to:

- Business requirements
- Retention policy
- Cost considerations
- Legal requirements

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17. Database Infrastructure

a. Managed Databases

Where practical, production systems should use managed database services.

Benefits include:

- Automated backups
- Monitoring
- Patch management
- Availability features
- Operational automation

b. Database Availability

Critical databases should have appropriate:

- Replication
- Backup
- Monitoring
- Recovery procedures

c. Database Access

Database access must use approved identity mechanisms.

Direct production database access should be restricted.

d. Database Performance

Database teams monitor:

- Query latency
- CPU
- Memory
- Storage
- Connections
- Replication health

Sustained resource utilization above **80%** should trigger capacity review for critical production databases.

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18. Backup Management

a. Backup Principles

Important systems must have appropriate backups.

Backup design should consider:

- Recovery requirements
- Data criticality
- Retention
- Cost
- Geographic resilience

b. Backup Frequency

Typical backup schedules may include:

Workload	Reference Frequency
Critical database	Continuous / frequent
Standard production database	Daily + transaction recovery where supported
File services	Daily
Configuration	Daily or version-controlled
Low-criticality systems	Weekly or according to requirements

c. Backup Verification

Backups are not considered reliable merely because they completed successfully.

Critical backups should be restored and tested periodically.

d. Backup Retention

Typical retention may include:

- Daily backups: 30 days
- Monthly backups: 12 months
- Annual archival: according to business/legal requirements

Actual retention depends on system requirements.

19. Disaster Recovery

a. Recovery Objectives

Systems may have defined:

- Recovery Time Objective (RTO)
- Recovery Point Objective (RPO)

Example classifications:

Criticality	Reference RTO	Reference RPO
Critical	4 hours	1 hour
High	8 hours	4 hours
Standard	24 hours	24 hours
Low	72 hours	48 hours

Actual customer contracts or business continuity requirements may establish stricter objectives.

b. Recovery Testing

Critical systems should participate in recovery testing at least **annually**.

High-criticality systems may require more frequent testing.

c. Recovery Documentation

Recovery documentation should identify:

- System owner
- Dependencies
- Recovery sequence
- Required credentials
- Infrastructure
- Validation steps
- Escalation contacts

---

20. Monitoring & Observability

a. Infrastructure Monitoring

Infrastructure monitoring should cover:

- CPU

- Memory
- Disk
- Network
- Availability
- Error rates
- Service health

b. Application Monitoring

Application teams should monitor:

- Requests
- Latency
- Errors
- Dependency failures
- Business transactions

c. Alerts

Alerts should be:

- Actionable
- Assigned to an owner
- Prioritized
- Documented

Teams should avoid alerts that consistently produce no action.

d. Alert Thresholds

Example thresholds:

Metric	Warning	Critical
CPU	70%	85%
Memory	75%	90%
Disk	70%	85%
Error rate	2%	5%
API p95 latency	700 ms	1,500 ms

Actual thresholds may differ according to workload characteristics.

21. Infrastructure Incident Management

a. Incident Classification

Infrastructure incidents may include:

- Cloud outage
- Network outage
- Database failure
- Storage failure
- Authentication failure
- Capacity exhaustion
- Configuration error
- Security event

b. Severity

Severity	Example
Sev 1	Critical enterprise/customer outage
Sev 2	Significant service degradation
Sev 3	Limited impact
Sev 4	Minor issue

c. Incident Process

```
Detection
   ↓
Triage
   ↓
Severity Assignment
   ↓
Mitigation
   ↓
Recovery
   ↓
Validation
   ↓
Root Cause Analysis
   ↓
Corrective Action
```

d. Incident Communication

For significant incidents, the incident lead should establish:

- Incident channel

- Technical owner
- Communications owner
- Status cadence
- Escalation path

22. Cloud Cost Management

a. Cost Ownership

Every production cloud project must have:

- Cost center
- Business owner
- Technical owner

b. Cost Monitoring

Cloud costs should be reviewed at least monthly.

Large or rapidly increasing costs should be investigated.

c. Cost Categories

Typical cloud costs include:

- Compute
- Storage
- Database
- Networking
- Data processing
- Security
- Monitoring
- AI/ML services

d. Budget Alerts

Cloud budgets should have alert thresholds.

A reference configuration is:

Threshold	Action
50%	Monitoring
75%	Owner notification
90%	Finance + technical owner review

Threshold	Action
100%	Budget investigation

e. Cost Optimization

Teams should consider:

- Right-sizing
- Autoscaling
- Reserved capacity
- Storage lifecycle policies
- Idle resource removal
- Query optimization

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23. Infrastructure as Code

a. Standard

Production infrastructure should be managed through version-controlled infrastructure-as-code where practical.

b. Benefits

Infrastructure-as-code provides:

- Repeatability
- Traceability
- Reviewability
- Automation
- Faster recovery

c. Change Process

Infrastructure changes should follow:

```
Code Change
   ↓
Review
   ↓
Validation
   ↓
Plan / Preview
   ↓
Approval
   ↓
Apply
   ↓
Verification
```

d. State Management

Infrastructure state must be protected from unauthorized access and accidental deletion.

---

24. Configuration Management

a. Configuration

Configuration should be separated from application code.

Examples:

- Environment variables
- Feature flags
- Service endpoints
- Resource limits
- Runtime settings

b. Sensitive Configuration

Sensitive configuration must use approved secret-management systems.

c. Configuration Changes

Material configuration changes affecting production should be traceable.

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25. Patch Management

a. Operating System Patches

Production operating systems should receive security updates according to vulnerability severity and operational risk.

b. Reference Targets

Severity	Reference Target
Critical	7 days
High	14 days
Medium	30 days
Low	90 days

Emergency security patches may require accelerated deployment.

c. Patch Exceptions

Exceptions must document:

- Vulnerability
- Affected system
- Reason for delay
- Compensating control
- Owner
- Target remediation date

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26. Capacity Management

a. Capacity Planning

Teams should monitor:

- CPU
- Memory
- Storage
- Network
- Database connections
- Queue depth
- API traffic

b. Capacity Threshold

A production service consistently operating above **80% of designed capacity** should be reviewed.

c. Growth Planning

Teams should consider expected growth over:

- 30 days
- 90 days
- 12 months

Critical infrastructure should not rely on emergency scaling as its normal capacity strategy.

---

27. Platform Engineering

a. Internal Platforms

Platform Engineering provides reusable capabilities to product teams.

Examples include:

- CI/CD templates
- Container platforms
- Logging
- Monitoring
- Identity integration
- Infrastructure modules
- Developer environments

b. Platform Standards

Internal platforms should prioritize:

- Self-service
- Security
- Reliability
- Documentation
- Consistency

c. Platform Service Levels

Critical platform services should define:

- Availability target
- Support model
- Escalation
- Maintenance windows

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28. Enterprise Integration

a. Integration Principles

Enterprise integrations should define:

- System owner
- Data owner
- Authentication
- Data exchanged
- Frequency
- Failure handling
- Monitoring

b. Integration Types

Nexora may use:

- REST APIs
- Events
- Message queues
- Batch files
- Managed integrations

c. Integration Failures

Integration systems should handle:

- Timeout
- Retry
- Duplicate messages
- Invalid data
- Authentication failure
- Provider outage

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29. Employee Connectivity

a. Office Networks

Corporate offices provide managed connectivity for:

- Employee devices
- Approved corporate services
- Conference systems

- Printers
- Guest access

b. Guest Networks

Guest networks should be logically separated from corporate systems.

c. VPN / Secure Remote Access

Remote access to internal systems must use approved secure-access mechanisms.

d. Network Performance

Corporate IT should monitor:

- Availability
- Internet bandwidth
- Latency
- Packet loss
- Access-point health

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30. Collaboration Technology

a. Collaboration Services

Nexora Workspace supports:

- Email
- Calendar
- Document collaboration
- Team communication
- Meetings

b. Access

Access should be provisioned based on employment status and role.

c. Data Protection

Sensitive company information must only be shared through approved collaboration services.

d. External Sharing

External sharing should be controlled according to information classification.

Restricted information should not be externally shared without authorization.

31. Endpoint Security

a. Endpoint Controls

Company-managed endpoints should use:

- Disk encryption
- Endpoint detection and response
- Central management
- Automatic security updates
- Screen locking
- Application controls

b. Security Baseline

A standard endpoint should:

- Lock after inactivity
- Require authentication
- Use supported operating systems
- Run approved security software

c. Personal Devices

Personal devices may only access corporate resources where specifically permitted.

Access may be restricted based on security and data classification requirements.

32. Software Licensing

a. License Management

Corporate IT maintains visibility of important software licenses.

b. Procurement

Software requiring payment must follow procurement procedures.

c. License Compliance

Employees must not:

- Install unauthorized commercial software
- Share license keys

- Circumvent license controls
- Use pirated software

d. License Optimization

Unused licenses should be reclaimed where practical.

Enterprise software utilization should be reviewed periodically.

---

33. Technology Procurement

a. Procurement Process

Technology procurement normally follows:

```
Business Requirement
       ↓
Technical Assessment
       ↓
Security Review
       ↓
Budget Confirmation
       ↓
Vendor Evaluation
       ↓
Commercial Negotiation
       ↓
Approval
       ↓
Purchase
       ↓
Implementation
```

b. Vendor Evaluation

Technology vendors may be evaluated on:

- Security
- Reliability
- Price
- Support
- Scalability
- Integration
- Contract terms

c. High-Risk Vendors

Vendors handling Restricted data or critical infrastructure may require enhanced security and legal review.

34. Maintenance Windows

a. Standard Maintenance

Maintenance should be planned to minimize business disruption.

b. Maintenance Communication

Planned maintenance should communicate:

- Date
- Time
- Expected impact
- Duration
- Services affected
- Contact

c. Emergency Maintenance

Emergency maintenance may be performed when required to protect:

- Availability
- Security
- Data integrity
- Customer impact

Emergency changes must be documented retrospectively.

35. Service Availability

a. Availability Targets

Reference targets for enterprise services include:

Service Class	Target
Critical	≥99.95%
High	≥99.9%
Standard	≥99.5%
Low	≥99.0%

Customer contracts may specify different targets.

b. Availability Calculation

Availability may be calculated as:

```
Availability =
(Total Scheduled Time - Downtime)
÷
Total Scheduled Time
× 100
```

For a 30-day month:

```
Total time = 720 hours
99.9% availability = approximately 43.2 minutes maximum downtime
```

Actual contractual calculations may exclude approved maintenance and defined exclusions.

---

36. Service Ownership & Operational Readiness

Before a new production service is launched, the service should have:

- Owner
- Technical team
- Support path
- Monitoring
- Alerting
- Documentation
- Deployment process
- Backup where applicable
- Recovery plan
- Security assessment

A service without operational ownership should not normally enter production.

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37. Infrastructure Change Management

a. Standard Changes

Standard, repeatable changes may follow pre-approved procedures.

Examples:

- Routine patching
- Approved certificate renewal

- Standard user provisioning

b. Normal Changes

Normal infrastructure changes require:

- Description
- Risk
- Implementation plan
- Validation
- Rollback
- Approval

c. Emergency Changes

Emergency changes are permitted when necessary to prevent or resolve significant incidents.

Emergency changes must be reviewed afterward.

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38. Cloud Security Controls

Cloud environments should use:

- Centralized identity
- MFA
- Role-based access
- Network segmentation
- Logging
- Monitoring
- Encryption
- Vulnerability scanning
- Security alerts

Public cloud resources must not be exposed unintentionally.

a. Public Resource Review

Public-facing storage, databases and management interfaces should receive additional review.

b. Encryption

Sensitive information should use approved encryption mechanisms both:

- At rest
- In transit

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39. Infrastructure Documentation

Each critical infrastructure service should maintain documentation covering:

- Architecture
- Ownership
- Dependencies
- Configuration
- Monitoring
- Backup
- Recovery
- Deployment
- Escalation

Documentation should be reviewed when major architecture changes occur.

40. Infrastructure Operating Metrics

Cloud and IT leadership may review:

Metric	Reference Target
Critical service availability	≥99.95%
High service availability	≥99.9%
Critical incident response	≤15 minutes
P1 ServiceDesk response	≤15 minutes
Standard CI pipeline	≤15 minutes
Critical vulnerability remediation	≤7 days
Critical access review	Quarterly
Critical backup recovery test	At least annually
Standard laptop replacement review	~36 months
Cloud cost review	Monthly
Critical service ownership	100%

41. Example Cloud Architecture

A typical Nexora production workload may use:



Supporting capabilities include:

- Centralized logging
- Monitoring
- Security monitoring
- Secret management
- CI/CD
- Backup

---

42. Example Production Environment

Environment: NXR-DATA-PROD-01

Business Function: Data & AI

Primary Region: Approved GCP region

Workloads:

- Data ingestion
- Transformation
- Analytics APIs
- Scheduled processing

Controls:

- MFA
- Role-based access
- Centralized logging

- Monitoring
- Backup
- Security scanning

Ownership:

- Technical Owner: Data Platform Engineering
- Infrastructure Owner: Cloud & Infrastructure
- Security Owner: Security Engineering
- Business Owner: Data & AI

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43. Example IT Service Request

Request ID: IT-2026-08421

Employee: NXR-EMP-02641

Request: Additional development monitor

Reason: Approved development workstation requirement

Priority: P4

Approval: Engineering Manager

Asset: NXR-MON-018274

Status: Fulfilled

Corporate IT records the assignment in the asset inventory.

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44. Example Infrastructure Incident

Incident ID: INC-2026-0317

Severity: Sev 2

Service: Customer Analytics API

Detection: Automated monitoring

Start: 14 July 2026, 14:08 IST

Initial symptom: API error rate increased from 0.4% to 7.2%.

Initial action:

1. Incident declared
2. Application team engaged
3. Infrastructure metrics reviewed
4. Database connection pool investigated

5. Traffic reduced
6. Service stabilized

Recovery: 14:31 IST

Total impact: 23 minutes

Follow-up:

- Root-cause analysis
- Connection-pool adjustment
- Capacity review
- Alert improvement

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45. Cloud Financial Management Example

Cloud Project: NXR-DATA-PROD-01

Cost Category	Monthly Example
Compute	₹1,850,000
Database	₹620,000
Storage	₹180,000
Data Processing	₹740,000
Networking	₹310,000
Monitoring/Security	₹210,000
Total	**₹3,910,000**

If monthly spending exceeds the approved budget by more than **10%**, the technical owner and Finance should review the variance.

Cost optimization opportunities may include:

- Instance right-sizing
- Storage lifecycle policies
- Query optimization
- Idle-resource removal
- Reserved capacity

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46. Infrastructure Capacity Example

A production API currently handles:

- 1,800 requests/second average
- 3,200 requests/second peak
- 68% average CPU
- 76% average memory
- 420 ms p95 latency

The team expects traffic to increase by approximately **35% over the next six months**.

The team should evaluate:

- Horizontal scaling
- Database capacity
- Cache capacity
- Network throughput
- Cost impact

Capacity planning should occur before the expected growth reaches existing limits.

---

47. Cross-Functional Responsibilities

Infrastructure operations frequently intersect with other Nexora functions.

Activity	Infrastructure	Other Function
Cloud architecture	Cloud	Engineering
Access management	Infrastructure	Security
Employee device	IT	People Operations
Software purchase	IT	Procurement
Cloud budget	Infrastructure	Finance
Customer environment	Infrastructure	Delivery
Data platform	Infrastructure	Data & AI
AI infrastructure	Infrastructure	AI Engineering
Incident response	Infrastructure	Security / Engineering
Disaster recovery	Infrastructure	Corporate Operations

The responsible owner must be identified for each production service.

---

48. Infrastructure Governance

a. Architecture Governance

Major infrastructure architecture decisions may require Technology Architecture Council review.

b. Security Governance

Infrastructure handling sensitive information must comply with Security requirements.

c. Financial Governance

Cloud spending must remain within approved budgets unless an approved variance exists.

d. Operational Governance

Critical infrastructure should participate in:

- Monitoring reviews
- Capacity reviews
- Backup testing
- Recovery exercises
- Incident reviews

---

49. Infrastructure Exceptions

Exceptions to this handbook must be documented.

An exception should include:

- Requirement
- Reason
- Risk
- Compensating control
- Owner
- Approval
- Expiry date

Temporary exceptions should include a remediation plan.

Security, legal and regulatory requirements cannot be bypassed through an ordinary infrastructure exception.

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50. Quick Reference

Topic	Standard
Primary cloud platform	Google Cloud Platform
Standard environments	Development, Test, Staging, Production
MFA	Required for corporate and privileged access
Critical access review	Quarterly
Standard laptop replacement review	~36 months
P1 IT response	15 minutes
P1 target resolution	4 hours
P2 IT response	30 minutes
Standard production logs	30 days
Security-relevant logs	90 days
Critical vulnerability target	7 days
High vulnerability target	14 days
Medium vulnerability target	30 days
Critical service availability	≥99.95%
High service availability	≥99.9%
Standard service availability	≥99.5%
Critical recovery test	At least annually
Cloud cost review	Monthly
Capacity review threshold	~80% sustained utilization
Secret rotation target	90 days

Appendix A — Standard Cloud Project Naming

```
NXR-<FUNCTION>-<ENVIRONMENT>-<NUMBER>
```

Examples:

```
NXR-DATA-PROD-01
NXR-DATA-STG-01
NXR-APP-PROD-02
NXR-SEC-PROD-01
NXR-PLATFORM-NONPROD-01
```

Appendix B — Standard Infrastructure Tags

Production cloud resources should use appropriate metadata.

Recommended fields include:

```
company = nexora
environment = production
business_unit = data-ai
application = analytics-platform
owner = data-platform
cost_center = CC-4207
classification = confidential
managed_by = cloud-infrastructure
```

Appendix C — Service Readiness Checklist

Before Production

- [ ] Owner assigned
- [ ] Business owner assigned
- [ ] Environment created
- [ ] Identity configured
- [ ] Network controls configured
- [ ] Secrets configured
- [ ] Monitoring enabled
- [ ] Logging enabled
- [ ] Backup configured
- [ ] Recovery requirements defined
- [ ] Security review completed
- [ ] Cost center assigned
- [ ] Deployment documented
- [ ] Support path defined

After Production

- [ ] Health validated
- [ ] Monitoring alerts tested
- [ ] Backup verified
- [ ] Access reviewed
- [ ] Cost monitored
- [ ] Documentation published

Appendix D — Backup and Recovery Matrix

System Class	Backup	Recovery Test	Reference RTO	Reference RPO
Critical	Continuous / frequent	Annual or more	4 hours	1 hour
High	Daily	Annual	8 hours	4 hours
Standard	Daily	Periodic	24 hours	24 hours
Low	Weekly	As required	72 hours	48 hours

Appendix E — IT ServiceDesk Priority Matrix

Priority	Example	Response	Resolution Target
P1	Enterprise-wide outage	15 min	4 hours
P2	Major team impact	30 min	8 hours
P3	Individual business impact	4 hours	2 business days
P4	Standard request	1 business day	5 business days

Appendix F — Related Documents

- **NXR-CORP-001** — Company, People & Corporate Governance Handbook
- **NXR-TECH-002** — Product, Engineering & Software Development Handbook
- **NXR-ENG-STD-003** — Production Change Management
- **NXR-SEC-POL-001** — Information Security Policy
- **NXR-SEC-STD-002** — Information Classification Standard
- **NXR-SEC-IRP-003** — Security Incident Response Plan
- **NXR-DATA-STD-001** — Data Governance Standard
- **NXR-AI-POL-001** — Responsible AI & Generative AI Policy

- **NXR-OPS-STD-001** — Business Continuity Standard
- **NXR-FIN-POL-001** — Financial Controls
- **NXR-LEG-POL-001** — Contract & Legal Review

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End of Document